



Distributed state estimation for jointly observable linear systems over time-varying networks[☆]

Shimin Wang, Martin Guay^{*}

Department of Chemical Engineering, Queen's University, Kingston, ON K7L 3N6, Canada

ARTICLE INFO

Article history:

Received 6 March 2023

Received in revised form 29 August 2023

Accepted 5 January 2024

Available online 18 February 2024

Keywords:

Time-varying networks

Distributed state estimation

Jointly observable systems

Linear time-invariant systems

ABSTRACT

This paper deals with a distributed state estimation problem for jointly observable multi-agent systems operated over various time-varying network topologies. The results apply when the system matrix of the system to be observed contains eigenvalues with positive real parts. They also can apply to situations where the communication networks are disconnected at every instant. We present sufficient conditions for the existence of distributed observers for general linear systems over periodic communication networks. Using an averaging approach, it is shown that the proposed distributed observer can provide exponentially converging state estimates of the state of the linear system when the network is uniformly connected on average. This average connectedness condition offers a more relaxed assumption that includes periodic switching, Markovian switching and Cox process switching as special cases. All the agents in the network share the estimated state with their neighbors through the network and cooperatively reconstruct the entire state locally. Furthermore, this study presents two exponential stability results for two classes of switched systems, providing valuable tools in related distributed state estimation approaches. A toy example and three practical applications are provided to illustrate the effectiveness of the theoretical results.

© 2024 Elsevier Ltd. All rights reserved.

1. Introduction

The cooperative control of complex systems has received tremendous attention in many areas of investigation, such as distributed sensor fusion in sensor networks (Dougherty & Guay, 2016; Olfati-Saber & Shamma, 2005) and unmanned aerial vehicle formations (Bar-Shalom & Li, 1995), amongst others. An important class of cooperative control problems is the leader–follower consensus problem (Priscoli, Isidori, Marconi, & Pietrabissa, 2015; Su & Huang, 2011). The objective of this problem is to design a distributed control system that synchronizes the state of the followers with the leader's state (Hong, Chen, & Bushnell, 2008). One major difficulty with this problem is that the state of the leader is generally not directly available for each follower or some followers. This problem requires one to design distributed observer techniques that enable state estimation in large-scale dynamical systems operating over a network of spatially distributed sensors (Liu & Huang, 2023). These difficulties from

cooperative control and practical demands have motivated the study of distributed state estimation problems, as originally described in Hong et al. (2008), Olfati-Saber (2007), Su and Huang (2011) and Su and Huang (2012) and fully developed in Han, Trentelman, Wang, and Shen (2018), Mitra and Sundaram (2018), Wang and Morse (2017) and Kim, Lee, and Shim (2019).

The distributed state estimation problem aims to design distributed observers for each agent (follower system or distributed sensor) to asymptotically reconstruct the state of the (leader or large-scale) system to be observed by using the local measurements and information obtained from its neighbors (Mitra & Sundaram, 2018; Su & Huang, 2011; Wang & Morse, 2017). The existing research addressing the distributed state estimation problem can be classified into the following four categories that reflect the observability assumptions of the system to be observed and the communication network: (1) Disjointly observable systems (Olfati-Saber, 2007), (2) Semi-jointly observable systems (Su & Huang, 2011, 2012), (3) Locally jointly observable systems (Mitra & Sundaram, 2018), and, (4) Jointly observable systems (Han et al., 2018; Kim et al., 2019; Wang & Morse, 2017; Yang, Barboni, Rezaee, & Parisini, 2022; Yang, Rezaee, Alessandri, & Parisini, 2023).

The disjoint observability assumption allows each agent to independently reconstruct the state of the system to be observed without interaction or communication, implying that the estimation of each agent does not impact the observability of other

[☆] This work was supported by the Natural Sciences and Engineering Research Council (NSERC), Canada. The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Vijay Gupta under the direction of Editor Christos G. Cassandras.

^{*} Corresponding author.

E-mail addresses: shimin.wang@queensu.ca (S. Wang), martin.guay@queensu.ca (M. Guay).

agents. A distributed algorithm was constructed in [Olfati-Saber \(2007\)](#) using a group of sensors over a static undirected graph under the disjoint observability assumption with cooperative terms to improve the performance of estimation. For semi-jointly observable systems, some special agents are equipped with sensors that lead to the observability of the full system to be observed. The rest of the agents that do not meet this observability condition require interactions with these special agents. The semi-jointly observable assumption has been employed for the solution of many leader-following consensus problems and cooperative tracking control problems ([Zhang & Lewis, 2012](#); [Zhang, Lewis, & Das, 2011](#)). It is worth mentioning that this assumption has also been used in [Su and Huang \(2011, 2012\)](#) who proposed a distributed observer scheme that solves a cooperative output regulation problem. The locally jointly observable assumption was initially proposed in [Mitra and Sundaram \(2018\)](#) to address situations where each agent and its connected neighbors collectively yield an output matrix that is detectable. The jointly observable assumption provides the mildest possible restriction on the system to be observed and the network. This assumption does not require the full observability of any agent in the network. It allows for the reconstruction of the system to be observed via local interactions among agents and local partial measurements.

The distributed state estimation problem over jointly observable assumption has been solved under various communication conditions for the linear system case in [Han et al. \(2018\)](#), [Kim et al. \(2019\)](#), [Wang, Liu, Morse, and Anderson \(2022\)](#), [Wang and Morse \(2017\)](#), [Yang et al. \(2022, 2023\)](#) and [Liu and Huang \(2023\)](#), and the nonlinear system case in [Wu, Isidori, and Lu \(2021\)](#). Specifically, a simple Luenberger-type local observer was presented in [Kim et al. \(2019\)](#) by using the Kalman observable canonical decomposition, which also promotes some variations of the distributed observer as in [Han et al. \(2018\)](#), [Jiao, Trentelman, and Camlibel \(2022\)](#), [Wang, Liu, et al. \(2022\)](#) and [Wang, Pan, and Guay \(2022\)](#) for the static communication network. For example, [Wang, Liu, et al. \(2022\)](#) propose a split-spectrum-based distributed estimator for the case when the system dynamics are continuous and the network is stationary. Moreover, state estimation for a class of linear time-invariant systems with unknown inputs has been considered in [Cao and Wang \(2023\)](#), [Yang et al. \(2022\)](#), which is a meaningful and practical consideration.

Static communication networks are an ideal condition. In real world applications, the communication networks can change dynamically due to various factors such as time-varying environment, communication distance, failures of links and network congestion. Some efforts have been made to investigate the distributed state estimation problem for continuous-time systems over time-varying graphs ([Liu & Huang, 2022](#); [Wang, Liu, & Morse, 2020](#); [Yang et al., 2023](#); [Zhang, Lu, Deng, & Chen, 2021](#)). The distributed estimation problems for switching networks that are strongly connected at every time with continuous-time and discrete-time systems were tackled in [Wang et al. \(2020\)](#), [Wang, Liu, et al. \(2022\)](#), respectively. In [Yang et al. \(2023\)](#), [Zhang et al. \(2021\)](#) and [Liu and Huang \(2022\)](#), jointly connected switching networks were investigated. This assumption constitutes the weakest requirement on the switching networks. It allows for networks that can be disconnected at every time instant. However, we note that the problem considered in [Liu and Huang \(2022\)](#), [Zhang et al. \(2021\)](#) and [Yang et al. \(2023\)](#) required the neutral stability of the matrix of the system to be observed, i.e. all the eigenvalues of the system matrix are semi-simple with zero real parts, or all the observer pairs (consisting of the system matrix and the output matrix) are marginally stabilizable, respectively.

This paper proposes the design of distributed observers for a general linear time-invariant system to estimate the state of

the system to be observed subject to jointly observable systems operated over time-varying graphs. All the agents in the network cooperate to reconstruct the state of the system to be observed and share the estimated state with their neighbors through local communication. The results allow the observed system to be unstable. The communication network can be disconnected at every instant. This class of networks can include connected static networks or every-time connected switching networks as special cases. Some sufficient conditions are presented for the existence of distributed observers for general linear systems to be observed over periodic communication networks. Using an averaging approach, it is shown that the distributed observer can estimate the state of the system to be observed exponentially fast over uniformly connected on average. This approach provides a more relaxed assumption on the network and can include periodic switching, Markovian switching and, Cox process switching, as special cases. Two exponential stability results are established for two classes of switched systems.

The rest of the paper is organized as follows. In Section 2, we introduce some standard assumptions and lemmas. Section 3 is devoted to the design and analysis of the proposed distributed observers over time-varying networks. This is followed by a simulation study in Section 4. Four simulation examples are presented which include one toy example and three practical applications. This is followed by brief conclusions in Section 5.

Notation: Let $\|\cdot\|$ denote both the Euclidean norm of a vector and the Euclidean-induced matrix norm (spectral norm) of a matrix. $\mathbb{R}$ is the set of real numbers. $\mathbb{N}$ denotes all natural numbers. I_n denotes the $n \times n$ identity matrix. For $A \in \mathbb{R}^{m \times n}$, $\text{Ker}(A) = \{x \in \mathbb{R}^n | Ax = 0\}$ and $\text{Im}(A) = \{y \in \mathbb{R}^m | y = Ax \text{ for some } x \in \mathbb{R}^n\}$ denote the kernel and range of A , respectively. For a subspace $\mathcal{V} \subset \mathbb{R}^n$, the orthogonal complement of $\mathcal{V}$ is denoted as $\mathcal{V}^\perp = \{x \in \mathbb{R}^n | x^T v = 0, \forall v \in \mathcal{V}\}$. $\otimes$ denotes the Kronecker product of matrices. For $b_i \in \mathbb{R}^{n_i \times p}$, $i = 1, \dots, m$, $\text{col}(b_1, \dots, b_m) \triangleq [b_1^T \dots b_m^T]^T$. For $a_i \in \mathbb{R}^{p \times n_i}$, $i = 1, \dots, m$, $\text{row}(a_1, \dots, a_m) \triangleq [a_1 \dots a_m]$. For $X_1 \in \mathbb{R}^{n_1 \times m_1}, \dots, X_k \in \mathbb{R}^{n_k \times m_k}$,

$$\text{diag}(X_1, \dots, X_k) \triangleq \begin{bmatrix} X_1 & & \\ & \ddots & \\ & & X_k \end{bmatrix}.$$

For a matrix $P \in \mathbb{R}^{n \times n}$, let λ_p and λ_P denote the minimum and maximum eigenvalues of P , respectively.

2. Problem formulation and assumptions

We consider a linear time-invariant system of the form:

$$\dot{x}(t) = Ax(t), \quad (1)$$

$$y_i(t) = C_i x(t), \quad i = 1, \dots, N \quad (2)$$

where $x(t) \in \mathbb{R}^n$ is the vector of state variables, and $y_i \in \mathbb{R}^{p_i}$ is the output of system (1) detected by agent i 's sensor, the known matrices A and C_i are of proper dimension, for $i = 1, \dots, N$.

As in [Zhang et al. \(2021\)](#) and [Liu and Huang \(2022\)](#), it is assumed that the multi-agent system is composed of N agents. The network topology of the multi-agent system is described by a switching digraph $\mathcal{G}_{\sigma(t)} = (\mathcal{V}, \mathcal{E}_{\sigma(t)})$ where $\sigma(t)$ is a switching signal with a dwelling time $\tau > 0$. For the vertices $\mathcal{V} = \{1, \dots, N\}$, we have that $(i, j) \in \mathcal{E}_{\sigma(t)}$ if and only if $a_{ji}(t) > 0$ at time instant t . We denote the set of neighbors of agent i at time t as $\mathcal{N}_i(t)$. For further details on graph theory, the reader is referred to [Zhang, Li, Qu, and Lewis \(2015\)](#).

The objective of this paper is to design distributed observers over time-varying networks to estimate the state of system (1) in

the sense that the estimation state $\hat{x}_i(t)$ of each agent's observer converges to the state $x(t)$, i.e.,

$$\lim_{t \rightarrow \infty} (\hat{x}_i(t) - x(t)) = 0, \quad i = 1, \dots, N.$$

Before we proceed, the following assumptions are needed.

Assumption 1. There exists a subsequence $\{i_k | k = 0, 1, \dots\}$ of $\{i | i = 0, 1, 2, \dots\}$ with $t_{i_{k+1}} - t_{i_k} < \nu$ for some positive ν such that the union graph $\bigcup_{j=i_k}^{i_{k+1}-1} \mathcal{G}_{\sigma(t_j)}$ is a strongly connected graph.

Assumption 2. The switching signal $\sigma(t)$ is periodic.

Remark 1. Under [Assumption 2](#), we can assume that the switching signal is as follows:

$$\sigma(t) = \begin{cases} 1, & \text{if } sT \leq t < (s + \omega_1)T; \\ 2, & \text{if } (s + \omega_1)T \leq t < (s + \sum_{\rho=1}^2 \omega_{\rho})T; \\ \vdots & \vdots \\ p, & \text{if } (s + \sum_{\rho=1}^{p-1} \omega_{\rho})T \leq t < (s + 1)T; \end{cases} \quad (3)$$

where T is a positive constant, $s = 0, 1, 2, \dots$, and $\omega_{\rho}, \rho = 1, \dots, p$ are positive constants satisfying $\sum_{\rho=1}^p \omega_{\rho} = 1$. Under [Assumptions 1](#) and [2](#), for any $k = 0, 1, 2, \dots$, the union graph $\bigcup_{j=i_k}^{i_{k+1}-1} \mathcal{G}_{\sigma(t_j)} = \bigcup_{\rho=1}^p \mathcal{G}_{\rho} \equiv \mathcal{G}$. Let $\mathcal{L}$ denote the Laplacian matrix associated with graph $\mathcal{G}$.

A more relaxed assumption is the so-called uniformly connected on average networks property ([Kim, Shim, Back, & Seo, 2013](#); [Stilwell, Boltt, & Roberson, 2006](#); [Wang & Nesic, 2010](#)). This assumption is shared by many time-varying networks with switching topologies such as periodic switching, Markovian switching and Cox process switching networks.

Assumption 3. A graph $\mathcal{G}_{\sigma(t)}$ with its corresponding Laplacian matrix $\mathcal{L}_{\sigma(t)}$, is said to be connected on average with uniform convergence to the average (for short, uniformly connected on average) if the average Laplacian

$$\mathcal{L} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_t^{t+T} \mathcal{L}_{\sigma(\tau)} d\tau, \quad t \geq 0.$$

induces a graph $\mathcal{G}$ that is connected.

Remark 2. The convergence of the limit is uniform with respect to t in the sense that there is a continuous and strictly decreasing function $\delta : [0, \infty)$, called a convergence function, such that $\lim_{T \rightarrow \infty} \delta(T) = 0$ and

$$\left\| \frac{1}{T} \int_t^{t+T} \mathcal{L}_{\sigma(\tau)} d\tau - \mathcal{L} \right\| \leq \delta(T), \quad t \geq 0.$$

Motivated by [Kim et al. \(2013\)](#), [Wang and Nesic \(2010\)](#) and [Stilwell et al. \(2006\)](#), we consider an averaging approach to describe the fast-switching network. We consider the time-varying Laplacian matrix of the form $\mathcal{L}_{\sigma(t/\epsilon)}$, where the parameter $\epsilon > 0$ determines the switching speed of the network topology.

Assumption 4. The system defined in (1) is jointly observable

Remark 3. For $i \in \mathcal{V}$, we assume that the observability index of (C_i, A) is v_i , such that $\text{rank}(\mathcal{O}_i) = v_i$, where $\mathcal{O}_i \in \mathbb{R}^{(\sum p_i) \times n}$ is the observability matrix given by: $\mathcal{O}_i = \text{col}(C_i, C_i A, \dots, C_i A^{n-1})$. For $i \in \mathcal{V}$, the observable subspace and unobservable subspace of (C_i, A) are defined as $\text{Im}(\mathcal{O}_i^T) \subset \mathbb{R}^n$ and $\text{Ker}(\mathcal{O}_i) \subset \mathbb{R}^n$, respectively, and satisfy $\text{Ker}(\mathcal{O}_i)^\perp = \text{Im}(\mathcal{O}_i^T)$.

For $i \in \mathcal{V}$, let $V_i = \text{row}(V_{ui}, V_{oi}) \in \mathbb{R}^{n \times n}$ be an orthogonal matrix such that $V_i V_i^T = I_n$. Let $V_{ui} \in \mathbb{R}^{n \times (n-v_i)}$ be a matrix such

that all columns of V_{ui} are from an orthogonal basis of the $\text{Ker}(\mathcal{O}_i)$ satisfying $\text{Im}(V_{ui}) = \text{Ker}(\mathcal{O}_i)$. Let $V_{oi} \in \mathbb{R}^{n \times v_i}$ be a matrix such that all columns of V_{oi} are from an orthogonal basis of the $\text{Im}(\mathcal{O}_i^T)$ satisfying $\text{Im}(V_{oi}) = \text{Im}(\mathcal{O}_i^T)$.

For $i \in \mathcal{V}$, we use the Kalman observability decomposition to express the matrices A and C_i of the system in (1) as follows:

$$V_i^T A V_i = \begin{bmatrix} A_{ui} & A_{ri} \\ 0_{v_i \times (n-v_i)} & A_{oi} \end{bmatrix}, \quad (4a)$$

$$C_i V_i = \begin{bmatrix} 0_{p_i \times (n-v_i)} & C_{oi} \end{bmatrix}, \quad (4b)$$

where the pair (C_{oi}, A_{oi}) is observable, $A_{oi} \in \mathbb{R}^{v_i \times v_i}$, $A_{ri} \in \mathbb{R}^{(n-v_i) \times v_i}$, $A_{ui} \in \mathbb{R}^{(n-v_i) \times (n-v_i)}$ and $C_{oi} \in \mathbb{R}^{p_i \times v_i}$ admit the following matrix decomposition: $A_{ui} = V_{ui}^T A V_{ui}$, $A_{ri} = V_{ui}^T A V_{oi}$, $A_{oi} = V_{oi}^T A V_{oi}$ and $C_{oi} = C_i V_{oi}$.

Let $C_0 = \text{diag}(C_{o1}, \dots, C_{oN})$, $A_r = \text{diag}(A_{r1}, \dots, A_{rN})$, $V_o = \text{diag}(V_{o1}, \dots, V_{oN})$, $V_u = \text{diag}(V_{u1}, \dots, V_{uN})$, $A_o = \text{diag}(A_{o1}, \dots, A_{oN})$ and $A_u = \text{diag}(A_{u1}, \dots, A_{uN})$.

Next, we state some useful lemmas proposed in [Kim et al. \(2019\)](#), [Sun and Ge \(2006\)](#) and [Qu \(2009\)](#)

Lemma 1 ([Sun and Ge \(2006\)](#)). Consider the linear switched system given by:

$$\dot{y} = A_{\sigma(t)} y \quad (5)$$

where $y \in \mathbb{R}^n$ is the vector of state variables, $\sigma : [0, \infty) \mapsto \mathcal{P} = \{1, 2, \dots, p\}$ is the switching signal satisfying [Assumption 2](#), and $A_{\rho} \in \mathbb{R}^{n \times n}$, $\rho = 1, \dots, p$. If the matrix $\sum_{\rho=1}^p \omega_{\rho} A_{\rho}$ is Hurwitz with ω_{ρ} , $\rho = 1, \dots, p$, obtained from (3), then there exists a positive constant $\bar{T}_0$ such that for $0 < T < \bar{T}_0$, the origin of system (5) is exponentially stable.

Lemma 2 ([Zhang et al. \(2015\)](#)). Suppose that the communication network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is strongly connected. Let $\theta = \text{col}(\theta_1, \dots, \theta_N)$ be the left eigenvector of the Laplacian matrix $\mathcal{L}$ associated with the eigenvalue 0, i.e., $\mathcal{L}^T \theta = 0$. Then, $\Theta = \text{diag}(\theta_1, \dots, \theta_N) > 0$ and $\hat{\mathcal{L}} = \Theta \mathcal{L} + \mathcal{L}^T \Theta \geq 0$.

Lemma 3 ([Kim et al. \(2019\)](#)). Suppose that the communication network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is strongly connected. Then, the following statements are equivalent:

1. System (1) is jointly observable;
2. The matrix $V_u^T (\hat{\mathcal{L}} \otimes I_n) V_u$ is positive definite;
3. The matrix $V_u^T (\mathcal{L} \otimes I_n) V_u$ is nonsingular.

Under [Assumptions 1](#) and [4](#), the matrix $V_u^T (\hat{\mathcal{L}} \otimes I_n) V_u$ is positive definite matrix from [Lemma 3](#). Let λ_l and λ_L denote the minimum and maximum eigenvalues of $V_u^T (\hat{\mathcal{L}} \otimes I_n) V_u$, respectively. Let $V_u = \text{diag}(V_{u1}, \dots, V_{uN})$ and $A_u = \text{diag}(A_{u1}, \dots, A_{uN})$. Then, we establish the following lemmas.

Lemma 4. Under [Assumption 4](#), consider the following linear time-invariant system

$$\dot{z} = [A_u - \gamma V_u^T (\mathcal{L} \otimes I_n) V_u] z, \quad (6)$$

where $z \in \mathbb{R}^{\sum_{i=1}^N (n-v_i)}$. Suppose that the communication network $\mathcal{G}$ is strongly connected. Then, the system (6) satisfies the following properties:

1. System (6) is globally asymptotically stable for a sufficiently large value of γ .
2. The matrix $A_u - \gamma V_u^T (\mathcal{L} \otimes I_n) V_u$ is Hurwitz for a sufficiently large value of γ .

Proof. Define the following Lyapunov function candidate for system (6)

$$V(z) = \sum_{i=1}^N \theta_i z_i^T z_i. \quad (7)$$

Then,

$$\theta_m \|z\|^2 \leq V(z) \leq \theta_M \|z\|^2, \quad (8)$$

where $\theta_m = \min\{\theta_1, \dots, \theta_N\}$ and $\theta_M = \max\{\theta_1, \dots, \theta_N\}$. The time derivative of $V(z)$ along (6) can be evaluated as

$$\begin{aligned} \dot{V} &= 2 \sum_{i=1}^N \theta_i z_i^T A_{ui} z_i - \gamma z^T [V_u^T (\hat{\mathcal{L}} \otimes I_n) V_u] z \\ &\leq 2\theta_M \|A\| \|z\|^2 - \gamma \lambda_l \|z\|^2. \end{aligned}$$

It is easily verified that $\|A_{ui}\| \leq \|A\|$, for $i = 1, \dots, N$. From (6) and (8), we have

$$\dot{V} \leq \frac{2\theta_M^2 \|A\| - \gamma \lambda_l \theta_m}{\theta_M \theta_m} V.$$

Hence, for any $\gamma > \frac{2\theta_M^2 \|A\|}{\lambda_l \theta_m}$, system (6) is globally asymptotically stable (exponentially stable). From the Theorem 4.5 in Khalil (2002), the matrix $A_u - \gamma V_u^T (\mathcal{L} \otimes I_n) V_u$ is Hurwitz for sufficiently large enough γ . $\square$

Lemma 5. Under Assumptions 1, 2 and 4, consider the following linear system

$$\dot{z} = [A_u - \gamma V_u^T (\mathcal{L}_{\sigma(t)} \otimes I_n) V_u] z, \quad (9)$$

where $z = \text{col}(z_1, \dots, z_N)$ with $z_i \in \mathbb{R}^{n-v_i}$. Then there exist some constants $\bar{\gamma}_0$ and $\bar{T}_0$, such that, for all $\gamma \geq \bar{\gamma}_0$ and $0 < T < \bar{T}_0$, the system (9) is exponentially stable at the origin.

Proof. Under Assumptions 1 and 2, for any $k = 0, 1, 2, \dots$, the union graph $\bigcup_{j=ik}^{ik+1-1} \mathcal{G}_{\sigma(t_j)} = \bigcup_{\rho=1}^p \mathcal{G}_{\rho} = \mathcal{G}$ is a strongly connected directed graph. Clearly, $\mathcal{L} = \sum_{\rho=1}^p \omega_{\rho} \mathcal{L}_{\rho}$. Then, from $\sum_{\rho=1}^p \omega_{\rho} = 1$, we have

$$\sum_{\rho=1}^p \omega_{\rho} [A_u - \gamma V_u^T (\mathcal{L}_{\rho} \otimes I_n) V_u] = A_u - \gamma V_u^T (\mathcal{L} \otimes I_n) V_u.$$

Thus, from Lemma 4, there exists $\bar{\gamma}_0$ such that $A_u - \gamma V_u^T (\mathcal{L} \otimes I_n) V_u$ is Hurwitz for any $\gamma \geq \bar{\gamma}_0$. Then, from Lemma 1, there exists a positive constant $\bar{T}_0$ such that for any $0 < T < \bar{T}_0$, system (9) is exponentially stable at the origin.

Lemma 6. Under Assumptions 1, 3 and 4, consider the following linear system

$$\dot{z} = [A_u - \gamma V_u^T (\mathcal{L}_{\sigma(t/\epsilon)} \otimes I_n) V_u] z, \quad (10)$$

where $z = \text{col}(z_1, \dots, z_N)$ with $z_i \in \mathbb{R}^{n-v_i}$. Suppose that the communication network $\mathcal{G}$ is strongly connected. Then, there exists a positive scalar ϵ^* such that, for any $0 < \epsilon < \epsilon^*$ and initial condition, the system (10) is exponentially stable at the origin. $\square$

Proof. The proof proceeds following the same steps as in Khalil (2002) and Kim et al. (2013). We define $Q_{av} = A_u - \gamma V_u^T (\mathcal{L} \otimes I_n) V_u$ and $Q(t/\epsilon) = \gamma V_u^T [(\mathcal{L} - \mathcal{L}_{\sigma(t/\epsilon)}) \otimes I_n] V_u$. Then, we have

$$\dot{z} = Q_{av} z + Q(t/\epsilon) z.$$

Under Assumption 3, and from Remark 2, $Q(t)$ satisfies

$$\left\| \frac{1}{T} \int_t^{t+T} Q(\tau) d\tau \right\| \leq k_1 \delta(T),$$

where $k_1 = \gamma \max_{i \in \mathcal{V}} \|V_{ui}\|^2$. Then, we define the following matrix

$$w(t, \epsilon) = \int_0^t Q(\tau) e^{-\epsilon(t-\tau)} d\tau,$$

for some positive constant ϵ . At $\epsilon = 0$, the function $w(t, 0)$ satisfies

$$\|w(t_1, 0) - w(t_2, 0)\| \leq k_1 |t_2 - t_1| \delta(t_2 - t_1), \quad (11)$$

where $k_1 = \gamma \max_{i \in \mathcal{V}} \|V_{ui}\|^2$. Integrating $w(t, \epsilon)$ by parts, we have

$$\begin{aligned} w(t, \epsilon) &= w(t, 0) - \epsilon \int_0^t e^{-\epsilon(t-\tau)} w(\tau, 0) d\tau \\ &= w(t, 0) - \epsilon \int_0^t e^{-\epsilon(t-\tau)} w(t, 0) d\tau \\ &\quad - \epsilon \int_0^t e^{-\epsilon(t-\tau)} [w(\tau, 0) - w(t, 0)] d\tau \\ &= w(t, 0) - w(t, 0) [e^{-\epsilon(t-\tau)}]_0^t \\ &\quad - \epsilon \int_0^t e^{-\epsilon(t-\tau)} [w(\tau, 0) - w(t, 0)] d\tau \\ &= w(t, 0) - w(t, 0) [1 - e^{-\epsilon t}] \\ &\quad - \epsilon \int_0^t e^{-\epsilon(t-\tau)} [w(\tau, 0) - w(t, 0)] d\tau \\ &= e^{-\epsilon t} w(t, 0) - \epsilon \int_0^t e^{-\epsilon(t-\tau)} [w(\tau, 0) - w(t, 0)] d\tau. \end{aligned} \quad (12)$$

Using the last expression, it follows that $w(t, \epsilon)$ fulfills the inequality:

$$\|w(t, \epsilon)\| \leq t e^{-\epsilon t} k_1 \delta(t) + k_1 \epsilon \int_0^t e^{-(t-\tau)\epsilon} (t - \tau) \delta(t - \tau) d\tau. \quad (13)$$

It follows from Khalil (2002) that there exists a class $\mathcal{K}$ function κ such that $\epsilon \|w(t, \epsilon)\| \leq \kappa(\epsilon)$ for all $s \leq 0$ and $0 \leq \epsilon \leq 1$. Therefore, $\epsilon \|w(t, \epsilon)\| = O(\kappa(\epsilon))$.¹ Let us define a new variable $\tilde{z}$ satisfying

$$z = [I + \epsilon w(t/\epsilon, \epsilon)] \tilde{z}.$$

As $\epsilon \|w(t/\epsilon, \epsilon)\| = O(\kappa(\epsilon))$ for all $t \geq 0$, there exists a constant $\epsilon_1 > 0$ such that $I + \epsilon w(t/\epsilon, \epsilon)$ is nonsingular for $0 \leq \epsilon \leq \epsilon_1$. Denote

$$[I + \epsilon w(t/\epsilon, \epsilon)]^{-1} \triangleq I + F(t/\epsilon, \epsilon).$$

It can be seen that $F(t/\epsilon, \epsilon) = O(\kappa(\epsilon))$ for all $0 \leq t$. The coordinate change yields the following dynamics

$$\begin{aligned} \dot{\tilde{z}} &= \epsilon \dot{w}(t/\epsilon, \epsilon) \tilde{z} + [I + \epsilon w(t/\epsilon, \epsilon)] \dot{\tilde{z}} \\ &= \epsilon \frac{d \left(\int_0^{t/\epsilon} Q(\tau) e^{-\epsilon(t/\epsilon-\tau)} d\tau \right)}{dt} \tilde{z} + [I + \epsilon w(t/\epsilon, \epsilon)] \dot{\tilde{z}} \\ &= \epsilon \left(Q(t/\epsilon)/\epsilon - \int_0^{t/\epsilon} Q(\tau) e^{-\epsilon(t/\epsilon-\tau)} d\tau \right) \tilde{z} + [I + \epsilon w(t/\epsilon, \epsilon)] \dot{\tilde{z}} \\ &= [Q(t/\epsilon) - \epsilon w(t/\epsilon, \epsilon)] \tilde{z} + [I + \epsilon w(t/\epsilon, \epsilon)] \dot{\tilde{z}}. \end{aligned} \quad (14)$$

Then, we have

$$\begin{aligned} \dot{\tilde{z}} &= [I + \epsilon w(t/\epsilon, \epsilon)]^{-1} \dot{\tilde{z}} \\ &\quad - [I + \epsilon w(t/\epsilon, \epsilon)]^{-1} [Q(t/\epsilon) - \epsilon w(t/\epsilon, \epsilon)] \tilde{z} \\ &= [I + \epsilon w(t/\epsilon, \epsilon)]^{-1} [Q_{av} + Q(t/\epsilon)] [I + \epsilon w(t/\epsilon, \epsilon)] \tilde{z} \\ &\quad - [I + \epsilon w(t/\epsilon, \epsilon)]^{-1} [Q(t/\epsilon) - \epsilon w(t/\epsilon, \epsilon)] \tilde{z} \end{aligned}$$

¹ $f_1(\epsilon) = O(f(\epsilon))$ if there are positive constants a and b such that $\|f_1(\epsilon)\| \leq a \|f(\epsilon)\|$ for $\|\epsilon\| < b$.

$$\begin{aligned}
&= [I + F(t/\epsilon, \epsilon)] [Q_{av} + Q(t/\epsilon)] [I + \epsilon w(t/\epsilon, \epsilon)] \tilde{z} \\
&\quad - [I + F(t/\epsilon, \epsilon)] [Q(t/\epsilon) - \epsilon w(t/\epsilon, \epsilon)] \tilde{z} \\
&= [I + F(t/\epsilon, \epsilon)] Q_{av} [I + \epsilon w(t/\epsilon, \epsilon)] \tilde{z} \\
&\quad + [I + F(t/\epsilon, \epsilon)] [Q(t/\epsilon) \tilde{z} + \epsilon Q(t/\epsilon) w(t/\epsilon, \epsilon) \tilde{z}] \\
&\quad - [I + F(t/\epsilon, \epsilon)] [Q(t/\epsilon) \tilde{z} - \epsilon w(t/\epsilon, \epsilon) \tilde{z}] \\
&= Q_{av} y + F(t/\epsilon, \epsilon) Q_{av} \tilde{z} \\
&\quad + \epsilon [I + F(t/\epsilon, \epsilon)] Q_{av} w(t/\epsilon, \epsilon) \tilde{z} \\
&\quad + \epsilon [I + F(t/\epsilon, \epsilon)] Q(t/\epsilon) w(t/\epsilon, \epsilon) \tilde{z} \\
&\quad + \epsilon [I + F(t/\epsilon, \epsilon)] w(t/\epsilon, \epsilon) \tilde{z} \\
&= [Q_{av} + N(t/\epsilon, \epsilon)] \tilde{z}, \tag{15}
\end{aligned}$$

where

$$\begin{aligned}
N(t/\epsilon, \epsilon) &= F(t/\epsilon, \epsilon) Q_{av} + \epsilon [I + F(t/\epsilon, \epsilon)] Q_{av} w(t/\epsilon, \epsilon) \\
&\quad + \epsilon [I + F(t/\epsilon, \epsilon)] [I + Q(t/\epsilon)] w(t/\epsilon, \epsilon).
\end{aligned}$$

By Lemma 4, it can be concluded that Q_{av} is Hurwitz for a sufficiently large value of γ . As a result, there exists a positive definite symmetric matrix P_{av} such that

$$P_{av} Q_{av} + Q_{av}^T P_{av} \leq -I.$$

We pose the following Lyapunov function candidate for system (15):

$$U(\tilde{z}) = \tilde{z}^T P_{av} \tilde{z}. \tag{16}$$

The time derivative of $U(\tilde{z})$ along the trajectories of (6) can be evaluated as

$$\begin{aligned}
\dot{U}(\tilde{z}) &= 2\tilde{z}^T P_{av} [Q_{av} + N(t/\epsilon, \epsilon)] \tilde{z} \\
&\leq -\tilde{z}^T \tilde{z} + 2\tilde{z}^T P_{av} N(t/\epsilon, \epsilon) \tilde{z} \\
&\leq -\|\tilde{z}\|^2 + 2\|P_{av}\| \|N(t/\epsilon, \epsilon)\| \|\tilde{z}\|^2. \tag{17}
\end{aligned}$$

It is noted that $N(t/\epsilon, \epsilon) = O(\kappa(\epsilon))$. Then, we have $\|N(t/\epsilon, \epsilon)\| \leq N^* \kappa(\epsilon)$ for some positive constant N^* . As a result, it follows from (17) that we have

$$\dot{U}(\tilde{z}) \leq -\left(1 - 2N^* \kappa(\epsilon) \|P_{av}\|\right) \|\tilde{z}\|^2.$$

Hence, for any $\epsilon \in (0, \epsilon^*)$ with $\epsilon^* = \min\left\{\kappa^{-1}\left(\frac{1}{2\|P_{av}\|N^*}\right), \epsilon_1\right\}$, we have $\dot{U}(\tilde{z}) < 0$. Therefore, the system (15) is exponentially stable at the origin for sufficiently small $\epsilon \in (0, \epsilon^*)$, together with $(Q_{av}z + Q(t/\epsilon)z)|_{z=0} \equiv 0$ for all $(t, \epsilon) \in [0, \infty) \times [0, \epsilon^*]$ further implies that (10) is exponentially stable at the origin for sufficiently small $\epsilon \in (0, \epsilon^*)$ from Theorem 10.5 in Khalil (2002). $\square$

3. Main results

3.1. Distributed observer over time-varying graphs

We now introduce the following linear dynamic observer:

$$\dot{\hat{x}}_i = A\hat{x}_i + L_i(C_i\hat{x}_i - y_i) + \gamma M_i \sum_{j \in N_i(t)} (\hat{x}_j - \hat{x}_i), \tag{18}$$

where, for $i = 1 \dots, N$, $\hat{x}_i \in \mathbb{R}^n$ is the estimate of x ,

$$L_i = V_i \begin{bmatrix} 0 \\ L_{oi} \end{bmatrix}, \quad M_i = V_i \begin{bmatrix} I_{n-v_i} & 0 \\ 0 & 0 \end{bmatrix} V_i^T,$$

γ is a sufficiently large positive constant to be determined and $L_{oi} \in \mathbb{R}^{v_i \times p_i}$ is chosen such that $(A_{oi} + L_{oi}C_{oi})$ is Hurwitz.

For $i = 1, \dots, N$, let $\tilde{x}_i = \hat{x}_i - x$ be the estimation error of the i th observer. Then, we have

$$\dot{\tilde{x}}_i = A\tilde{x}_i + L_i C_i \tilde{x}_i + \gamma M_i \sum_{j \in N_i(t)} (\tilde{x}_j - \tilde{x}_i)$$

$$= (A + L_i C_i) \tilde{x}_i - \gamma M_i \sum_{j=1}^N l_{ij}(t) \tilde{x}_j, \tag{19}$$

where $l_{ij}(t)$ is the (i, j) th entry of the Laplacian matrix $\mathcal{L}_{\sigma(t)}$ at time moment t . Let $\tilde{x}_{oi} = V_{oi}^T \tilde{x}_i$ and $\tilde{x}_{ui} = V_{ui}^T \tilde{x}_i$, for $i = 1, \dots, N$. Then, we have the following system from (4) and (19),

$$\begin{aligned}
\dot{\tilde{x}}_{ui} &= A_{ui} \tilde{x}_{ui} + A_{ri} \tilde{x}_{oi} \\
&\quad - \gamma V_{ui}^T \sum_{j=1}^N l_{ij}(t) [V_{uj} \tilde{x}_{uj} + V_{oj} \tilde{x}_{oj}], \tag{20a}
\end{aligned}$$

$$\dot{\tilde{x}}_{oi} = (A_{oi} + L_{oi} C_{oi}) \tilde{x}_{oi}. \tag{20b}$$

Let $\tilde{x}_u = \text{col}(\tilde{x}_{u1}, \dots, \tilde{x}_{uN})$, $\tilde{x}_o = \text{col}(\tilde{x}_{o1}, \dots, \tilde{x}_{oN})$, $V_o = \text{diag}(V_{o1}, \dots, V_{oN})$, and $A_r = \text{diag}(A_{r1}, \dots, A_{rN})$. Then, the system (20a) can be put into the following compact form,

$$\begin{aligned}
\dot{\tilde{x}}_u &= A_u \tilde{x}_u + A_r \tilde{x}_o - \gamma V_u^T (\mathcal{L}_{\sigma(t)} \otimes I_n) [V_u \tilde{x}_u + V_o \tilde{x}_o] \\
&= M(t) \tilde{x}_u + N(t) \tilde{x}_o \tag{21}
\end{aligned}$$

where $M(t) = A_u - \gamma V_u^T (\mathcal{L}_{\sigma(t)} \otimes I_n) V_u$ and $N(t) = A_r - \gamma V_u^T (\mathcal{L}_{\sigma(t)} \otimes I_n) V_o$.

Then, we have the following results.

Lemma 7. Under Assumption 4, consider system (1) and the linear switched system (18). Then,

$$\lim_{t \rightarrow \infty} (\hat{x}_i(t) - x(t)) = 0, \quad i = 1, \dots, N,$$

for any $x(0) \in \mathbb{R}^n$ and $\hat{x}_i(0) \in \mathbb{R}^n$, provided that the following system

$$\dot{\tilde{x}}_u = M(t) \tilde{x}_u, \tag{22}$$

is exponentially stable.

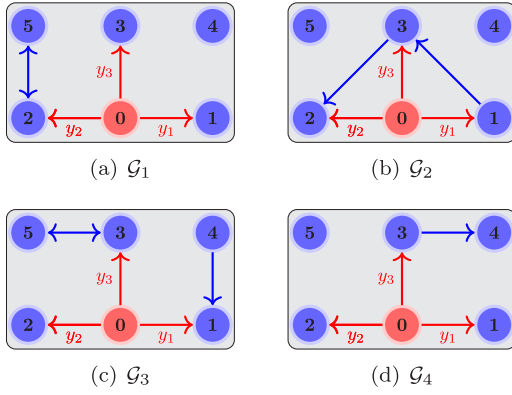
Proof. As $\tilde{x}_o(t) = 0$ for all $t \geq 0$, system (21) reduces to system (22). Since the system (22) is exponentially stable, it follows that, for any positive definite matrix $Q(t)$ satisfying $\|Q(t)\| \geq c_3$, for some positive constant c_3 , there exists a positive definite matrix $P(t)$ satisfying $c_1 \leq \|P(t)\| \leq c_2$ for some positive constants c_1 and c_2 which solves the matrix differential equation:

$$\dot{P}(t) = -P(t)M(t) - M^T(t)P(t) - Q(t). \tag{23}$$

We pose the following Lyapunov function candidate $V(\tilde{x}_u) = \tilde{x}_u^T P(t) \tilde{x}_u$. Its rate of change along the trajectories of system (21) is such that

$$\begin{aligned}
\dot{V} &= \tilde{x}_u^T [\dot{P}(t) + P(t)M(t) + M^T(t)P(t) \tilde{x}_u + 2P(t)N(t) \tilde{x}_o] \\
&= -\tilde{x}_u^T Q(t) \tilde{x}_u + 2\tilde{x}_u^T P(t) N(t) \tilde{x}_o \\
&\leq -c_3 \|\tilde{x}_u\|^2 + 2c_2 \|N(t)\| \|\tilde{x}_u\| \|\tilde{x}_o\| \\
&\leq -c_3 \|\tilde{x}_u\|^2 + \left[\frac{c_3}{4} \|\tilde{x}_u\|^2 + \frac{4c_2^2}{c_3} \|N(t)\|^2 \|\tilde{x}_o\|^2 \right] \\
&\leq -\frac{3c_3}{4} \|\tilde{x}_u\|^2 + \frac{4c_2^2}{c_3} \|N(t)\|^2 \|\tilde{x}_o\|^2 \\
&\leq -\frac{3c_3}{4c_2} V + \frac{4c_2^2}{c_3} \|N(t)\|^2 \|\tilde{x}_o\|^2. \tag{24}
\end{aligned}$$

From Assumption 4, L_{oi} was chosen such that $A_{oi} + L_{oi}C_{oi}$ is Hurwitz, for $i = 1, \dots, N$, and, therefore, we conclude that $\lim_{t \rightarrow \infty} \tilde{x}_{oi}(t) = 0$. As a result, we can also conclude that $\lim_{k \rightarrow \infty} \|\tilde{x}_o\|^2 = 0$. We note that the matrix $\mathcal{L}_{\sigma(t)}$ is a piecewise constant matrix where $\sigma(t)$ has the range $\mathcal{P} = \{1, \dots, p\}$. Thus, $N(t)$ is bounded over $[0, +\infty)$ and continuous on each time interval $[t_s, t_{s+1})$, $s = 0, 1, \dots$. It follows that, for all $t \geq 0$, $\lim_{t \rightarrow \infty} \frac{4c_2^2}{c_3} \|N(t)\|^2 \|\tilde{x}_o(t)\|^2 = 0$ exponentially. Hence, system

Fig. 1. Communication topology $\mathcal{G}_{\sigma(t)}$.

(23) can be viewed as an input to state stable system with $\frac{4c_2^2}{c_3} \|N(t)\|^2 \|\tilde{x}_o(t)\|^2$ as the input. Since this input tends to zeros, it follows that we can conclude that $\lim_{t \rightarrow \infty} V(t) = 0$ exponentially, which further implies $\lim_{t \rightarrow \infty} \tilde{x}_o(t) = 0$. Hence, $\lim_{t \rightarrow \infty} \tilde{x}_{ui}(t) = 0$ and $\lim_{t \rightarrow \infty} \tilde{x}_{oi}(t) = 0$. And, since, $\tilde{x}_{oi}(t) = V_{oi}^T \tilde{x}_i(t)$ and $\tilde{x}_{ui}(t) = V_{ui}^T \tilde{x}_i(t)$, we conclude that $\lim_{t \rightarrow \infty} \tilde{x}_i(t) = 0$, for $i = 1, \dots, N$. $\square$

Theorem 1. Consider system (1) and the linear switched system (18). Under Assumptions 1, 2 and 4, there exists positive $\bar{\gamma}_0 > 0$ and $\bar{T}_0$ such that for any $\gamma \geq \bar{\gamma}_0$ and $0 < T < \bar{T}_0$, $\lim_{t \rightarrow \infty} (\hat{x}_i(t) - x(t)) = 0$, exponentially for any $x(0) \in \mathbb{R}^n$ and $\hat{x}_i(0) \in \mathbb{R}^n$, $i = 1, \dots, N$.

Proof. In order to analyze system (21), we first assume that $\tilde{x}_o(t) = 0$ for all $t \geq 0$, such that system (21) will reduce to the system

$$\dot{\tilde{x}}_u = (A_u - \gamma V_u^T (\mathcal{L}_{\sigma(t)} \otimes I_n) V_u) \tilde{x}_u. \quad (25)$$

System (25) is in the form of (9). Under Assumptions 1, 2 and 4, from Lemma 5, there exist positive $\bar{\gamma}_0 > 0$ and $\bar{T}_0$ such that for any $\gamma \geq \bar{\gamma}_0$ and $0 < T < \bar{T}_0$, (25) is exponentially stable at the origin. Therefore, by using Lemma 7, we have $\lim_{t \rightarrow \infty} (\hat{x}_i(t) - x(t)) = 0$, exponentially for any $x(0) \in \mathbb{R}^n$ and $\hat{x}_i(0) \in \mathbb{R}^n$, $i = 1, \dots, N$. $\square$

Theorem 2. Consider system (1) and the linear switched system (18). Under Assumptions 3 and 4, there exists a positive scalar ε^* , for any $0 < \varepsilon < \varepsilon^*$ and initial condition, such that $\lim_{t \rightarrow \infty} (\hat{x}_i(t) - x(t)) = 0$, exponentially for any $x(0) \in \mathbb{R}^n$ and $\hat{x}_i(0) \in \mathbb{R}^n$, $i = 1, \dots, N$.

By using Lemmas 6 and 7, we can follow the proof of Theorem 1 to establish the result of Theorem 2. The proof is omitted.

4. Numerical example

4.1. Example 1: Toy example

In this example, we consider a linear distributed system composed of one leader system and five followers as shown in Fig. 1. The dynamic of the leader is in the form (1) with

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}, \quad B = 0, \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} C_1 \\ C_2 \\ C_3 \\ C_4 \\ C_5 \end{matrix}.$$

Next, we assume that the switching network topology $\bar{\mathcal{G}}_{\sigma(t)}$ is dictated by the following switching signal:

$$\sigma(t) = \begin{cases} 1 & \text{If } sT \leq t < (s + \frac{1}{4})T \\ 2 & \text{If } (s + \frac{1}{4})T \leq t < (s + \frac{1}{2})T \\ 3 & \text{If } (s + \frac{1}{2})T \leq t < (s + \frac{3}{4})T \\ 4 & \text{If } (s + \frac{3}{4})T \leq t < (s + 1)T \end{cases}$$

where $s = 0, 1, 2, \dots$. The four digraphs $\bar{\mathcal{G}}_k$, $k = 1, 2, 3, 4$, are illustrated in Fig. 1, where the node 0 is associated with the exosystem, and the other nodes are associated with the five followers.

It can be seen that none of the pairs (C_i, A) are observable, while the system to be observed is such that (C, A) is observable. Choosing the following matrix based on (4)

$$V_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad V_2 = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix},$$

$$V_3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad V_4 = I_3, \quad V_5 = I_3.$$

The partition y_1 , y_2 and y_3 of the output can be measured by agents 1, 2 and 3 as shown in Fig. 1 that satisfies Assumption 1 with

$$\mathcal{L} = \sum_{i=1}^4 \mathcal{L}_i = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 2 & -1 & 0 & -1 \\ -1 & 0 & 2 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 0 & 1 \end{bmatrix}.$$

Let $\theta = 10^{-3} \times \text{col}(1, 1, 1, 1, 2)$. Hence, $\lambda_L = 1.05 \times 10^{-2}$ and $\lambda_I = 1.3 \times 10^{-3}$. Thus, we can design a control law composed of Eq. (18) with the following parameters: $\gamma = 45$, $L_{o1} = \text{col}(-4 - 2)$, $L_{o2} = \text{col}(-4 - 2)$, $L_{o3} = -2.5$, $L_4 = \text{col}(0, 0, 0)$, $L_5 = \text{col}(0, 0, 0)$, $M_4 = I_3$ and $M_5 = I_3$. Simulation is conducted with the following initial conditions: $x(0) = \text{col}(1, 2, 3)$ and $\hat{x}_i(0) = \text{col}(0, 0, 0)$.

Fig. 2 shows the estimation errors of all nodes over different T . The results demonstrate that the estimation error approaches the estimation error of the averaged estimate as the frequency of switching is increased. Fig. 3 shows the estimation performance of each agent's state over $T = 0.1$.

4.2. Example 2: Applications to quarter-car active automotive suspension system

We borrow a modified example from Yue, Butsuen, and Hedrick (1988) and Sferlazza, Tarbouriech, and Zaccarian (2021) to illustrate the design of local observers over a time-varying network. Consider the quarter-car active automotive suspension system described in Fig. 4, which represents the automotive system at each wheel and is monitored by two separate sensors. The system consists of a spring k_s , a damper b_s and an active force actuator F_a . The sprung mass m_s represents the quarter-car equivalent of the vehicle body mass. The unsprung mass m_u represents the equivalent mass due to the axle and tire. The spring k_t represents the vertical stiffness of the tire. The variables z_s , z_u and z_r are the vertical displacements from static equilibrium of the sprung mass, unsprung mass and the road, respectively.

The dynamics of this system are governed by the following system

$$\dot{x}_c = A_c x_c + B_u F_a + B_d \dot{z}_r \quad (26)$$

where F_a is the active force of the actuator, $\dot{z}_r$ is an input describing how the road profile enters into the system, the state vector $x_c = \text{col}(x_{c1}, x_{c2}, x_{c3}, x_{c4})$ include the suspension deflection

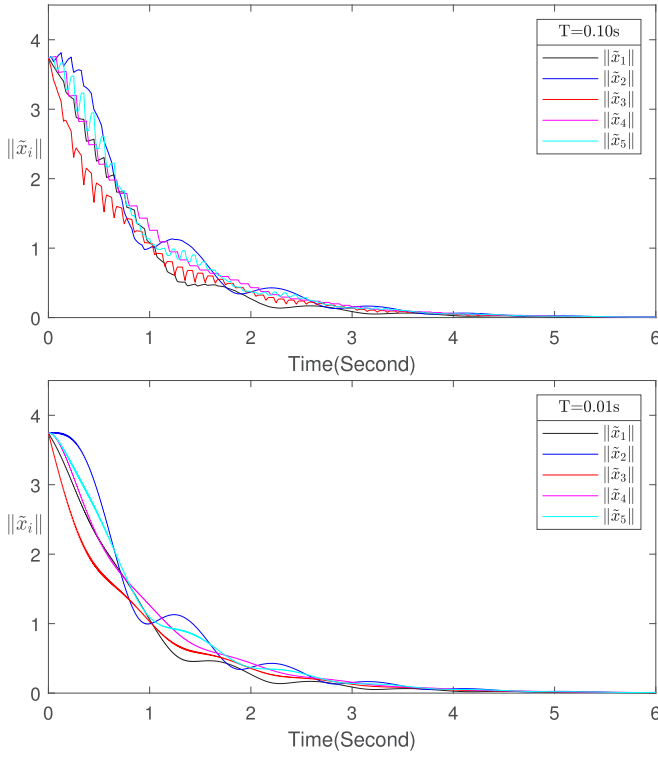


Fig. 2. Estimation errors of all nodes with different T , $i = 1, \dots, 5$.

$x_{c1} = z_s - z_u$, the absolute velocity $x_{c2} = \dot{z}_s$ of the sprung mass m_s , the tire deflection $x_{c3} = z_u - z_r$ and the absolute velocity $x_{c4} = \dot{z}_u$ of the unsprung mass m_u . The matrices A_c , B_u and B_d are given as

$$A_c = \begin{bmatrix} 0 & 1 & 0 & -1 \\ \frac{-k_s}{m_s} & \frac{-b_s}{m_s} & 0 & \frac{b_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_u} & \frac{b_s}{m_u} & \frac{-k_t}{m_u} & \frac{b_s+b_t}{-m_u} \end{bmatrix}, \quad B_u = \text{col}(0, \frac{1}{m_s}, 0, \frac{-1}{m_s}), \quad B_d = \text{col}(0, 0, -1, \frac{b_t}{m_u}).$$

where $m_s = 240$ Kg, $m_u = 36$ Kg, $b_s = 980$ N s/m, $k_s = 1.6 \times 10^4$ N/m, $k_t = 1.6 \times 10^5$ N/m and $b_t = 0$.

The time-varying networks are shown in Fig. 5. The network structure is driven by the switching signal:

$$\sigma(t) = \begin{cases} 1 & \text{If } sT \leq t < (s + \frac{1}{2})T \\ 2 & \text{If } (s + \frac{1}{2})T \leq t < (s + 1)T \end{cases}$$

where $s = 0, 1, 2, \dots$. We assume that the road profile can be viewed as a disturbance described by the following equation

$$\dot{v} = Sv, \quad (27a)$$

$$\dot{z}_r = Fv, \quad (27b)$$

where $v \in \mathbb{R}^4$ is the state of the exosystem, $F = [2 \ -1 \ -1 \ 2]$ and $S = \text{diag}(2a, a)$ are the output matrix and system matrix of the exosystem with $a = [\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}]$. We define $x = \text{col}(x_c, v)$ and rewrite (26) and (27) in the following compact form:

$$\dot{x} = Ax + BF_a \quad (28)$$

where $A = \begin{bmatrix} A_c & F \\ 0_{4 \times 4} & S \end{bmatrix}$ and $B = \begin{bmatrix} B_u \\ \text{hellocheckthecommand}_4 \end{bmatrix}$. It is also assumed that the active force actuator is $F_a = 0$. The variable $y_2 = C_2x$ is measured by agent 2 with $C_2^T = \text{col}(1, 0, 0, 0, 0_4)$. Agent 1 does not measure anything from the quarter-car active automotive suspension system. For this configuration, we get that $\text{rank}(\mathcal{O}_1) = 0$ and $\text{rank}(\mathcal{O}_2) = 8$. Furthermore, we confirm that (C, A) is observable with $C = \text{col}(C_1, C_2)$. Using the decomposition (4), we compute the transformed matrices and choose

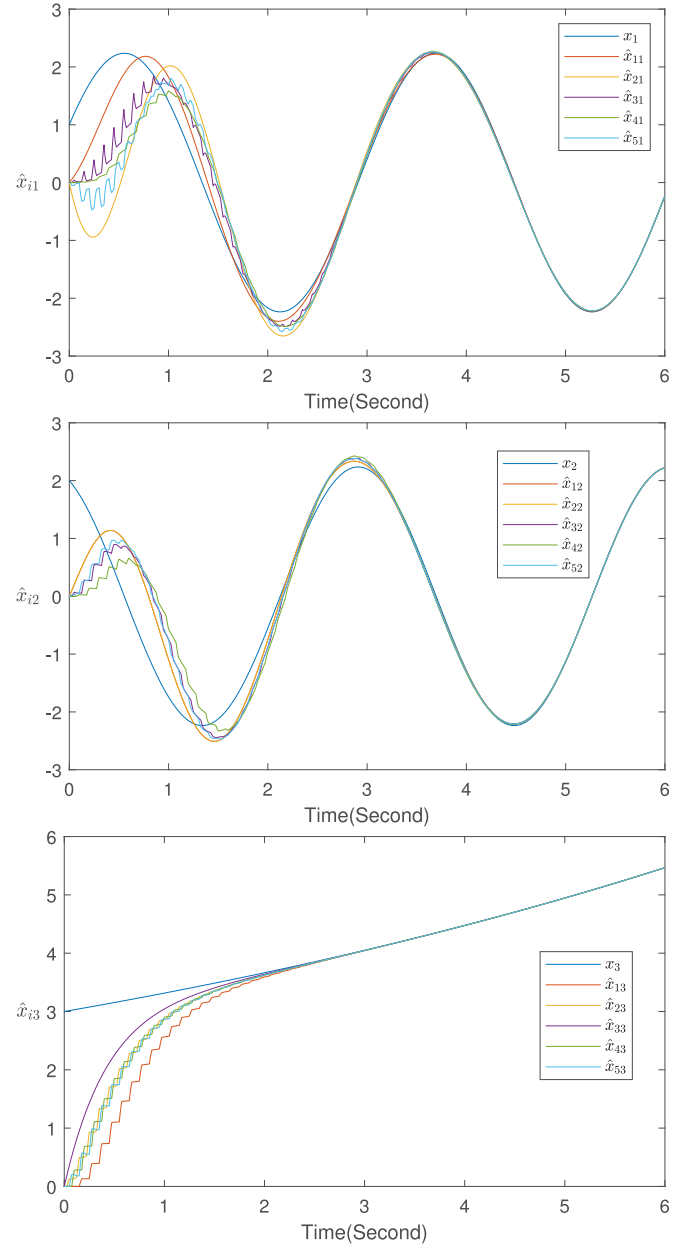


Fig. 3. Estimation performance of all nodes with $T = 0.1$, $i = 1, \dots, 5$.

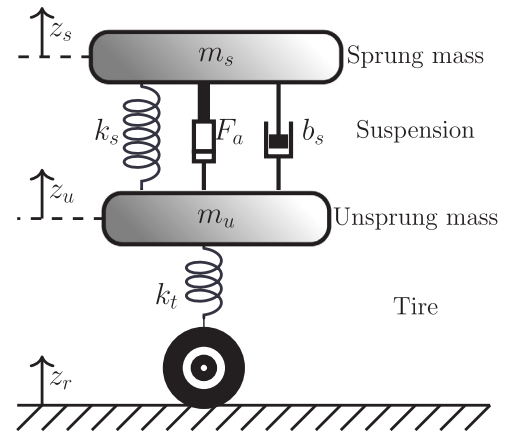
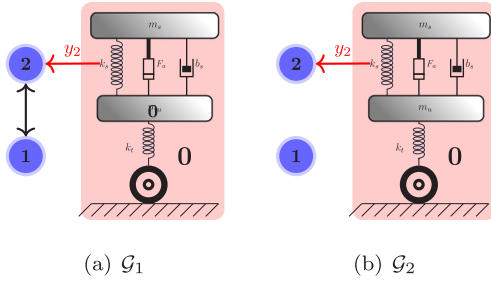
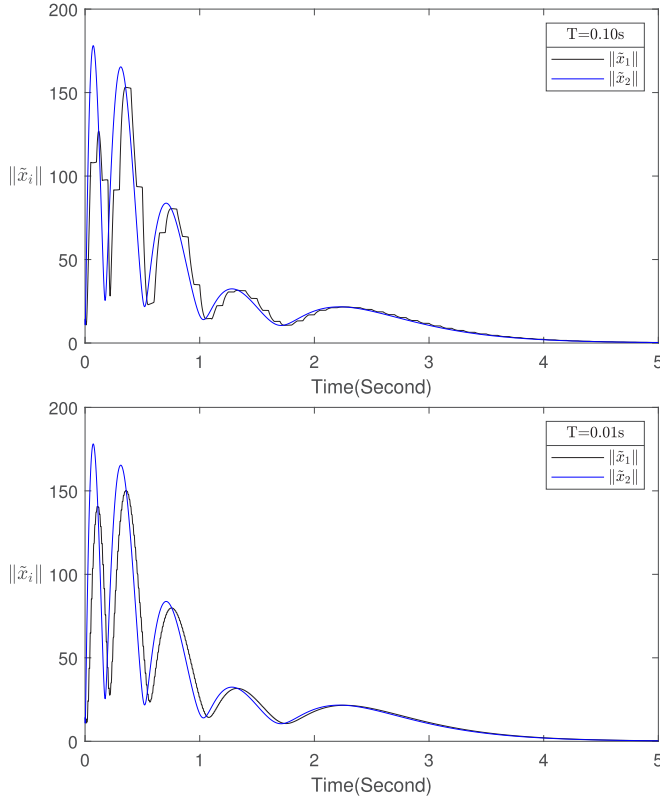


Fig. 4. Quarter-car active automotive suspension system.

Fig. 5. Communication topology and measurements $\mathcal{G}_{\sigma(t)}$.Fig. 6. Estimation errors of all nodes with different T .

$\gamma = 25$ and the vector L_{o2} such that the matrix $(A_{o2} + L_{o2}C_{o2})$ at $\{-2; -4; -6; -8; -3; -9; -5; -7\}$. The initial conditions for the simulation are generated randomly. Fig. 6 shows the estimation errors of all agents. It can be observed that agent 2 can reconstruct the state of (26). It also demonstrates that the distributed observer with fast-switching communication enables the estimation for the remote agent 1 which can approximate the averaged system.

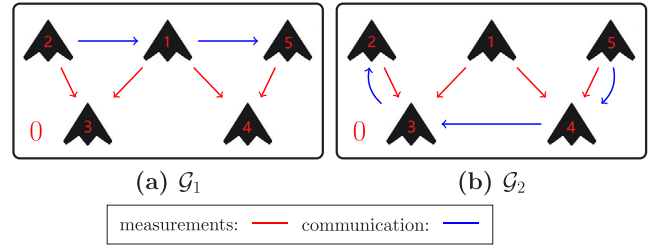
4.3. Example 3: Applications to the formation of unicycle-type vehicles

We now consider formation control of 5 flying unicycle-type vehicles with the i th agent's dynamics given by:

$$\dot{p}_{xi} = v_i \cos(\theta_i), \quad \dot{p}_{yi} = v_i \sin(\theta_i), \quad \dot{\theta}_i = \omega_i \quad (29)$$

where $p_i = \text{col}(p_{xi}, p_{yi}) \in \mathbb{R}^2$ is the position along the X- and Y-axis, θ_i is the attitude, v_i is the translational speed, and ω_i is the angular velocity of the i th vehicle, respectively.

As in Kim et al. (2019), we assume that the formation flight is conducted in unforced conditions for which v_i and ω_i are

Fig. 7. Communication networks $\mathcal{G}_{\sigma(t)}$ of flying vehicles.

constants. Let $v_i = \text{col}(v_{xi}, v_{yi})$ represent the velocity of the i th vehicle along the X- and Y-axis. Using the dynamics (29), we can establish that the dynamics of each vehicle are governed by the following two systems:

$$\begin{bmatrix} \dot{\theta}_i \\ \dot{\omega}_i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_i \\ \omega_i \end{bmatrix}, \quad \begin{bmatrix} \dot{p}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} 0_{2 \times 2} & I_2 \\ 0_{2 \times 2} & \omega_i a \end{bmatrix} \begin{bmatrix} p_i \\ v_i \end{bmatrix},$$

where $a = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. We assume that each vehicle has knowledge of its altitude and angular velocity ω_i . Each vehicle has only limited measurement ability. For example, as shown in Fig. 7, each vehicle can access its own position and velocity. However, only agents 3 and 4 can measure their relative position to other agents. The communication network structure and the local measurements for the network of vehicles are shown in Fig. 7. We define $x = \text{col}(p_1, v_1, \dots, p_5, v_5) \in \mathbb{R}^{20}$. The network of multi-vehicles can be written in the form of system (1) with

$$A = \text{diag}(A_1, A_2, A_3, A_4, A_5), \quad C_1 = [1 \ 0 \ 0 \ 0 \ 0] \otimes S, \\ C_2 = [0 \ 1 \ 0 \ 0 \ 0] \otimes S, \quad C_3 = [1 \ 1 \ 1 \ 0 \ 0] \otimes S, \\ C_4 = [1 \ 0 \ 0 \ 1 \ 1] \otimes S, \quad C_5 = [0 \ 0 \ 0 \ 0 \ 1] \otimes S,$$

where $A_i = \begin{bmatrix} 0_{2 \times 2} & I_2 \\ 0_{2 \times 2} & \omega_i a \end{bmatrix}$ and $S = \begin{bmatrix} I_2 & 0_{2 \times 2} \end{bmatrix}$ are the system and output matrix of the i th vehicle, C_i reflects the measurement ability of the i th vehicle in terms of other vehicles, A is system matrix of the system to be observed. We can also verify that $\text{rank}(C_1) = 4$, $\text{rank}(C_2) = 4$, $\text{rank}(C_3) = 4$, $\text{rank}(C_4) = 4$ and $\text{rank}(C_5) = 4$. As a result, no vehicle can observe the global information despite the fact that the pair (C, A) is observable with $C = \text{col}(C_1, C_2, C_3, C_4, C_5)$. Using the proposed observer, each vehicle can establish distributed observer to estimate the global information x over the time-varying networks shown in Fig. 7 with switching signal:

$$\sigma(t) = \begin{cases} 1 & \text{if } sT \leq t < (s + \frac{1}{2})T \\ 2 & \text{if } (s + \frac{1}{2})T \leq t < (s + 1)T \end{cases}$$

where $s = 0, 1, 2, \dots$

As above, the tuning parameters are chosen using the Kalman decomposition (4) for different values of ω_i . The simulation starts with angular velocity $\omega_i = 0$. At the time $t = 6$ s, the angular velocities of the vehicles are changed from $\omega_i = 0$ to $\omega_i = 0.5$. Fig. 8 shows the estimated positions and positions of all followers. Fig. 9 shows the estimation errors of all followers. We compute the transformed matrices and choose $\gamma = 25$ and the vector L_{oi} such that the matrix $(A_{oi} + L_{oi}C_{oi})$ has eigenvalues at $\{-1, -2, -3, -4\}$.

4.4. Example 4: Applications to position estimation in vehicular platoons of multiple connected and automated vehicles (CAV)

We now apply the distributed observer to the vehicular platoons control of multiple connected and automated vehicles with

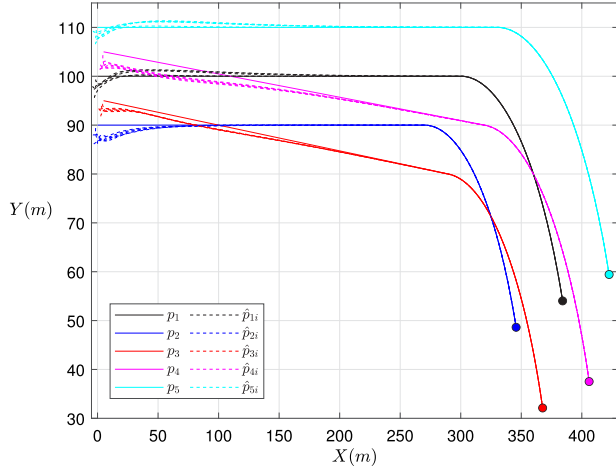
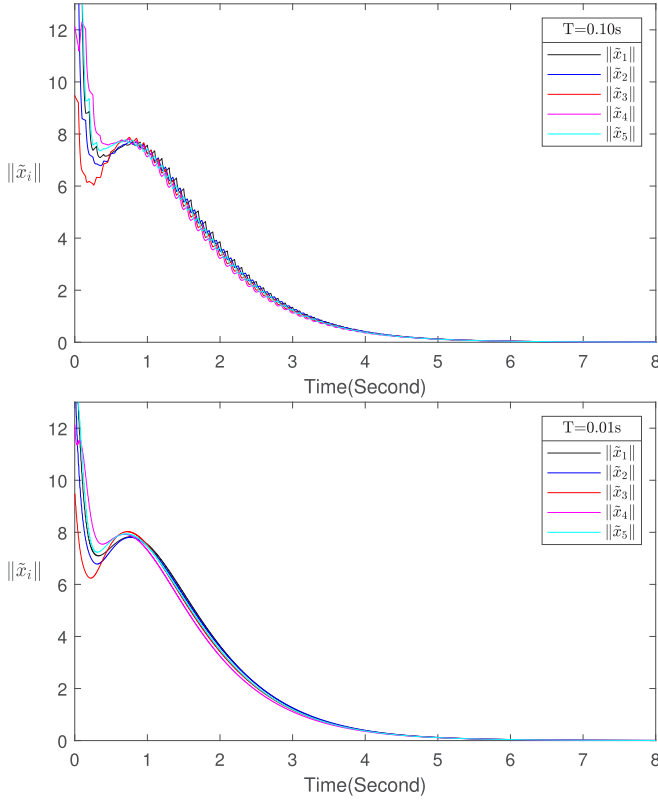


Fig. 8. Position and estimated position of all vehicles.

Fig. 9. Estimation errors of all vehicles with different T .

dynamics given by

$$\begin{bmatrix} \dot{p}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} 0_{2 \times 2} & I_2 \\ 0_{2 \times 2} & 0_{2 \times 2} \end{bmatrix} \begin{bmatrix} p_i \\ v_i \end{bmatrix} + \begin{bmatrix} 0_{2 \times 1} \\ u_i \end{bmatrix},$$

where $p_i = \text{col}(p_{xi}, p_{yi}) \in \mathbb{R}^2$ and $v = \text{col}(v_{xi}, v_{yi}) \in \mathbb{R}^2$ are the position and velocity of the i th CAV along the X - and Y -axis, respectively; $u_i \in \mathbb{R}^2$ is the control input of the i th CAV along the X - and Y -axis.

Each vehicle has limited measurement capability. As shown in Fig. 10, each vehicle can access its own position and velocity, but only a vehicle following another one can measure its relative position to the vehicle ahead. The network of all vehicles, their local measurements and communication network structure

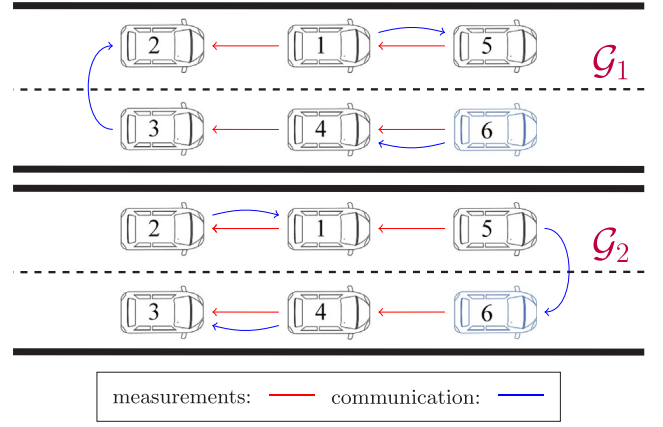
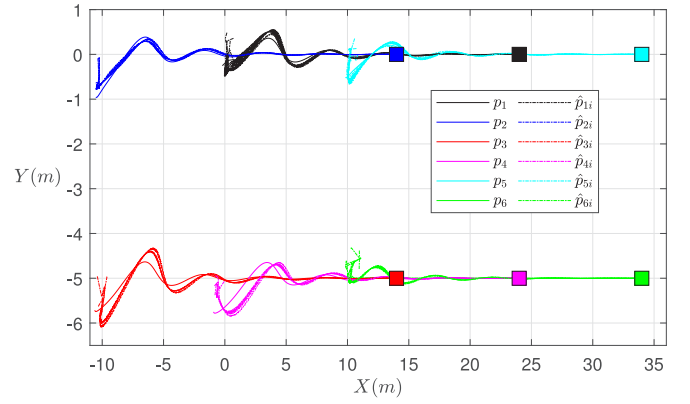
Fig. 10. Connected autonomous vehicles' communication networks $\mathcal{G}_{\sigma(t)}$.

Fig. 11. Position and Estimated position of all vehicles.

is shown in Fig. 10. In this case, the state of the network is denoted by $x = \text{col}(p_1, v_1, \dots, p_6, v_6) \in \mathbb{R}^{24}$ with dynamics (1) governed by the system matrix $A = I_5 \otimes \begin{bmatrix} 0_{2 \times 2} & I_2 \\ 0_{2 \times 2} & 0_{2 \times 2} \end{bmatrix}$, and with local measurement matrices $C_1 = [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0] \otimes S$, $C_2 = [1 \ 1 \ 0 \ 0 \ 0 \ 0] \otimes S$, $C_3 = [0 \ 0 \ 1 \ 1 \ 0 \ 0] \otimes S$, $C_4 = [0 \ 0 \ 0 \ 1 \ 0 \ 1] \otimes S$, $C_5 = [0 \ 0 \ 0 \ 0 \ 1 \ 0] \otimes S$ and $C_6 = [0 \ 0 \ 0 \ 0 \ 0 \ 1] \otimes S$, where $S = [I_2 \ 0_{2 \times 2}]$. Each C_i reflects the measurement ability of the i th vehicle.

We can also verify that $\text{rank}(\mathcal{O}_1) = 4$, $\text{rank}(\mathcal{O}_2) = 4$, $\text{rank}(\mathcal{O}_3) = 4$, $\text{rank}(\mathcal{O}_4) = 4$, $\text{rank}(\mathcal{O}_5) = 4$, $\text{rank}(\mathcal{O}_6) = 4$ and $\text{rank}(\mathcal{O}) = 24$ with $\mathcal{C} = \text{col}(C_1, C_2, C_3, C_4, C_5, C_6)$. As a result, no vehicle can reconstruct the state of other vehicles without coordination.

Suppose that vehicle 6 leads the vehicular network and that the following reference signal describes its position:

$$p_6(t) = \text{col}(v_{6x}t + p_{6x}, v_{6y}t + p_{6y}).$$

The control objective is to control each vehicle such that

$$\lim_{t \rightarrow \infty} (p_i(t) - p_6(t)) = p_{i6}$$

where $p_{i6} \in \mathbb{R}^2$ are the relative positions between i th vehicle and vehicle 6. The control input is chosen as

$$u_i = K(x_i - \hat{x}_{6i} - \text{col}(p_{i6}, 0, 0))$$

where $K^T = \text{col}(-8, -4) \otimes I_2$, $x_i = \text{col}(p_i, v_i)$ and $\hat{x}_{6i} = \text{col}(\hat{p}_{i6}, \hat{v}_{i6})$ is the vector of estimates of the states variables of vehicle 6 for the i th vehicle. We compute the transformed matrices using the Kalman decomposition (4), and choose $\gamma = 250$ and the vector L_{oi} such that the matrix $(A_{oi} + L_{oi}C_{oi})$ at $\{-1, -2, -3, -4\}$. Fig. 11

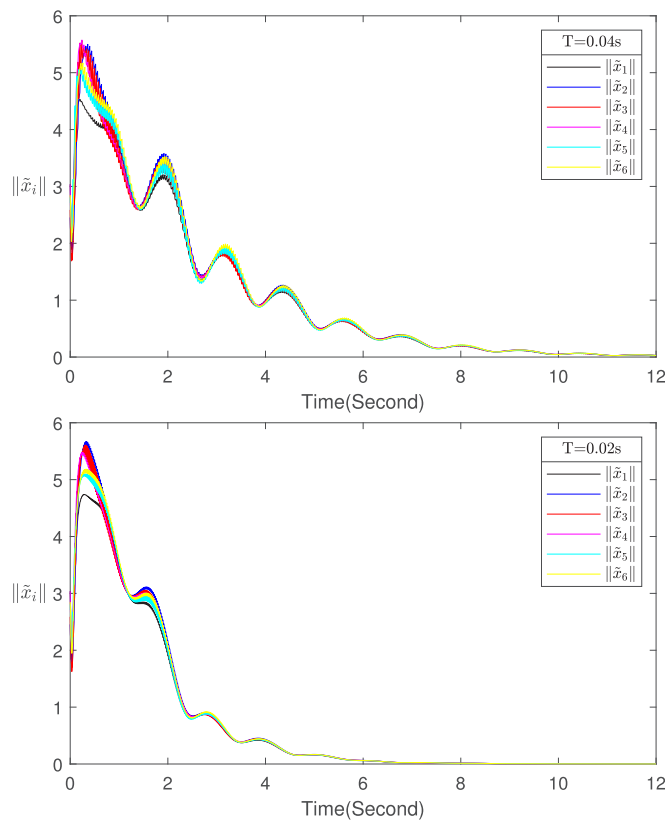


Fig. 12. Estimation errors of all vehicles with different T .

shows the estimated positions and the positions of all followers. Fig. 12 shows the estimation errors of all followers. Each vehicle has limited measurement ability and cannot measure the position of all the cars ahead. As we can see from Fig. 11, the use of the time-varying communication networks enables the reconstruction of the state of all vehicles from all vehicles in the network Fig. 10.

5. Conclusion

A distributed observer for a general linear time-invariant system is proposed to estimate the state of the system over time-varying jointly observable networks. An averaging approach is proposed to facilitate the analysis of the distributed observer scenarios that meet an average uniformly connected assumption. This assumption is met by a wide range of possible switching protocols, including periodic switching, Markovian switching and Cox process switching. It is shown that the distributed observer provides exponential convergence of the state estimates for each agent. The study provides two exponential stability results for two classes of switched systems, providing valuable tools for future research in related distributed state estimation fields.

References

- Bar-Shalom, Y., & Li, X.-R. (1995). *Multitarget-multisensor tracking: Principles and techniques*: vol. 19, YBs Storrs, CT.
- Cao, G., & Wang, J. (2023). A distributed reduced-order unknown input observer. *Automatica*, 155, Article 111174.
- Dougherty, S., & Guay, M. (2016). An extremum-seeking controller for distributed optimization over sensor networks. *IEEE Transactions on Automatic Control*, 62(2), 928–933.
- Han, W., Trentelman, H. L., Wang, Z., & Shen, Y. (2018). A simple approach to distributed observer design for linear systems. *IEEE Transactions on Automatic Control*, 64(1), 329–336.

- Hong, Y., Chen, G., & Bushnell, L. (2008). Distributed observers design for leader-following control of multi-agent networks. *Automatica*, 44(3), 846–850.
- Jiao, J., Trentelman, H. L., & Camlibel, M. K. (2022). H_2 and H_∞ suboptimal distributed filter design for linear systems. *IEEE Transactions on Automatic Control*, <http://dx.doi.org/10.1109/TAC.2022.3184399>.
- Khalil, H. K. (2002). *Nonlinear systems* (third ed.). Upper Saddle River, NJ: Patience Hall.
- Kim, T., Lee, C., & Shim, H. (2019). Completely decentralized design of distributed observer for linear systems. *IEEE Transactions on Automatic Control*, 65(11), 4664–4678.
- Kim, H., Shim, H., Back, J., & Seo, J. H. (2013). Consensus of output-coupled linear multi-agent systems under fast switching network: Averaging approach. *Automatica*, 49(1), 267–272.
- Liu, T., & Huang, J. (2022). Distributed exponential state estimation of linear systems over jointly connected switching networks. *arXiv preprint arXiv:2205.00218*.
- Liu, T., & Huang, J. (2023). Distributed exponential state estimation for discrete-time linear systems over jointly connected switching networks. *IEEE Transactions on Automatic Control*, <http://dx.doi.org/10.1109/TAC.2023.3241783>.
- Mitra, A., & Sundaram, S. (2018). Distributed observers for LTI systems. *IEEE Transactions on Automatic Control*, 63(11), 3689–3704.
- Olfati-Saber, R. (2007). Distributed Kalman filtering for sensor networks. In *2007 46th IEEE conference on decision and control* (pp. 5492–5498). IEEE.
- Olfati-Saber, R., & Shamma, J. S. (2005). Consensus filters for sensor networks and distributed sensor fusion. In *Proceedings of the 44th IEEE conference on decision and control* (pp. 6698–6703). IEEE.
- Priscoli, F. D., Isidori, A., Marconi, L., & Pietrabissa, A. (2015). Leader-following coordination of nonlinear agents under time-varying communication topologies. *IEEE Transactions on Control of Network Systems*, 2(4), 393–405.
- Qu, Z. (2009). *Cooperative control of dynamical systems: applications to autonomous vehicles*. London, U.K.: Springer Science & Business Media.
- Sferlaza, A., Tarbouriech, S., & Zaccarian, L. (2021). State observer with Round-Robin aperiodic sampled measurements with jitter. *Automatica*, 129, Article 109573.
- Stilwell, D. J., Boltt, E. M., & Roberson, D. G. (2006). Sufficient conditions for fast switching synchronization in time-varying network topologies. *SIAM Journal on Applied Dynamical Systems*, 5(1), 140–156.
- Su, Y., & Huang, J. (2011). Cooperative output regulation of linear multi-agent systems. *IEEE Transactions on Automatic Control*, 57(4), 1062–1066.
- Su, Y., & Huang, J. (2012). Cooperative output regulation with application to multi-agent consensus under switching network. *IEEE Transactions on Systems, Man and Cybernetics, Part B (Cybernetics)*, 42(3), 864–875.
- Sun, Z., & Ge, S. S. (2006). *Switched linear systems: control and design*. Longdon: Springer Science & Business Media.
- Wang, L., Liu, J., & Morse, A. S. (2020). A distributed observer for a continuous-time linear system with time-varying network. *arXiv preprint arXiv:2003.02134*.
- Wang, L., Liu, J., Morse, A. S., & Anderson, B. B. (2022). Distributed State Estimation for Linear Systems. *arXiv preprint arXiv:2210.01232*.
- Wang, L., & Morse, A. S. (2017). A distributed observer for a time-invariant linear system. *IEEE Transactions on Automatic Control*, 63(7), 2123–2130.
- Wang, W., & Nesic, D. (2010). Input-to-state stability and averaging of linear fast switching systems. *IEEE Transactions on Automatic Control*, 55(5), 1274–1279.
- Wang, S., Pan, Y., & Guay, M. (2022). Distributed state estimation for linear time-invariant systems with aperiodic sampled measurement. *arXiv preprint arXiv:2211.05223*.
- Wu, Y., Isidori, A., & Lu, R. (2021). On the design of distributed observers for nonlinear systems. *IEEE Transactions on Automatic Control*, 67(7), 3229–3242.
- Yang, G., Barboni, A., Rezaee, H., & Parisini, T. (2022). State estimation using a network of distributed observers with unknown inputs. *Automatica*, 146, Article 110631.
- Yang, G., Rezaee, H., Alessandri, A., & Parisini, T. (2023). State estimation using a network of distributed observers with switching communication topology. *Automatica*, 147, Article 110690.
- Yue, C., Butsuen, T., & Hedrick, J. (1988). Alternative control laws for automotive active suspensions. In *American control conference* (no. 25), (pp. 2373–2378).
- Zhang, H., & Lewis, F. L. (2012). Adaptive cooperative tracking control of higher-order nonlinear systems with unknown dynamics. *Automatica*, 48(7), 1432–1439.
- Zhang, H., Lewis, F. L., & Das, A. (2011). Optimal design for synchronization of cooperative systems: State feedback, observer and output feedback. *IEEE Transactions on Automatic Control*, 56(8), 1948–1952.
- Zhang, H., Li, Z., Qu, Z., & Lewis, F. L. (2015). On constructing Lyapunov functions for multi-agent systems. *Automatica*, 58, 39–42.
- Zhang, L., Lu, M., Deng, F., & Chen, J. (2021). Distributed state estimation of linear systems under uniformly connected switching networks. In *2021 60th IEEE conference on decision and control* (pp. 4008–4013). IEEE.



Shimin Wang received the B.Sci. degree in Mathematics and Applied Mathematics and M.Eng. degree in Control Science and Control Engineering from Harbin Engineering University, Harbin, China, in 2011 and 2014, respectively. He then received the Ph.D. degree in Mechanical and Automation Engineering from The Chinese University of Hong Kong, Hong Kong, SAR, China, in 2019.

He was a recipient of the NSERC Postdoctoral Fellowship award in 2022. From 2014 to 2015, he was an assistant engineer at the Jiangsu Automation Research Institute, China State Shipbuilding Corporation Limited. From 2019 to 2023, he held post-doctoral positions in the Department of Electrical and Computer Engineering at the University of Alberta and the Department of Chemical Engineering at Queen's University, Canada, respectively. He is now a postdoctoral associate at the Department of Chemical Engineering at Massachusetts Institute of Technology, USA.



Martin Guay received the Ph.D. degree from Queen's University, Kingston, ON, Canada, in 1996. He is currently a Professor in the Department of Chemical Engineering at Queen's University. His current research interests include nonlinear control systems, especially extremum-seeking control, nonlinear model predictive control, adaptive estimation and control, and geometric control.

He was a recipient of the Syncrude Innovation Award, the D. G. Fisher from the Canadian Society of Chemical Engineers, and the Premier Research Excellence Award. He is a Senior Editor of IEEE Control Systems Letters. He is the Editor-in-Chief of the Journal of Process Control. He is also an Associate Editor for Automatica, the IEEE Transactions on Automatic Control and the Canadian Journal of Chemical Engineering.